

GCDTRANSFORM

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Problem

Three positive integers are written on a blackboard. In each move, the numbers are first assigned the labels a, b and c in a way that $a > \gcd(b, c)$ holds, and then a is replaced with $a - \gcd(b, c)$. Show that the game always ends and always reaches the same three numbers $x \leq y \leq z$ for the same starting numbers.

Solution. Call one transformation to be a "move". Let $d = \gcd(a, b, c)$. Let a_n, b_n, c_n be the numbers after n moves.

Observe that $d \mid a_n, b_n, c_n$ for all n . Since we transform $a_{n-1} \rightarrow a_{n-1} - \gcd(b_{n-1}, c_{n-1}) = a_n$ with $a_{n-1} > \gcd(b_{n-1}, c_{n-1})$, we have $a_n, b_n, c_n > 0$ for all n .

Let $S_n = a_n + b_n + c_n$. Since after each move, the value of the number that the move is performed on decreases by at least 1, S_n is monotonically decreasing. Since we have $S_n \geq 3d$ for all n , the game must stop after a certain point.

Fix a, b, c . We will show that the game stops when eventually $a_n = b_n = c_n = d$ and that this state is attained.

If the game stops with $a_n = b_n = c_n$ not holding true, there can be two cases:

- $a_n > b_n \geq c_n$: Since $\gcd(b_n, c_n) \leq b_n < a_n$, so, $a_n \rightarrow a_{n+1}$ is a valid move.
- $a_n \geq b_n > c_n$: Since $\gcd(b_n, c_n) \leq \frac{b_n}{2}$, so, $a_n \rightarrow a_{n+1}$ is a valid move.

As a result, the game stops when $a_n = b_n = c_n$. We have, $\gcd(a_n, b_n, c_n) = a_n$.

Due to Bezout's Theorem, for any b_n, c_n , $\exists x_n, y_n$ such that $b_n x_n + c_n y_n = \gcd(b_n, c_n)$. After each move gets performed, $\gcd(a_n, b_n, c_n) = \gcd(a_n + b_n x_n + c_n y_n, b_n, c_n) = \gcd(a_{n+1}, b_{n+1}, c_{n+1})$. As a result the gcd is invariant. Thus, we must have $a_n = d$ which concludes the claim.

The last 3 numbers (x, y, z) are (d, d, d) .